

UKRAINIAN MILITARY RESEARCH: AI, ADAPTIVE EW, RF SIGNAL ANALYSIS & RELATED TECHNOLOGY

Technical synthesis of relevant developments in the 2025 Kharkiv National Air Force University conference proceedings

Prepared specifically to support analysis of the SIH problem statement: “Development of Smart Scan Strategy for Electronic Warfare...”

1. Source, Scope and Evidence Standard

Primary source: the official 2025 proceedings of the XXI International Scientific Conference of Ivan Kozhedub Kharkiv National Air Force University, “Новітні технології – для захисту повітряного простору” (“New Technologies – for the Protection of Aerospace”), held 9–10 April 2025. The university publication is 888 pages.

This report is deliberately focused on material that is technically relevant to your SIH problem: electronic warfare, RF/radio-electronic sensing, machine learning, adaptive detection, prediction, resource allocation/optimization, autonomous EW platforms, radar adaptation and related information-processing methods. It is NOT a claim to summarize every one of the hundreds of unrelated papers in the 888-page proceedings.

Evidence rule: a proposal described by the conference authors is labelled as a proposed/research approach. The report does not convert a conference proposal into a claim of operational deployment or demonstrated battlefield performance.

2. Conference Context

The proceedings state that they contain abstracts of theoretical and practical results of scientific research and development performed by university faculty, researchers, doctoral researchers, postgraduate researchers, military-management specialists, institutions and enterprises. The conference covers 25 sections.

Section	Printed start page	Relevance to SIH
Section 7 – Radio-technical troops / radar reconnaissance	258	Radar sensing, detection and interference environment
Section 8 – Military electronic communications / radio-technical support	305	RF communications and radio-electronic systems
Section 9 – Information and automated control technologies	346	Software, automation and information processing
Section 13 – Intelligence / special operations / search and rescue	510	Sensing, reconnaissance and information collection
Section 14 – Radio-electronic and cyber warfare	532	Directly relevant: EW, adaptive RF concepts, ML
Section 15 – Modern directions in radio electronics	569	Relevant enabling RF/electronics technologies

3. Most Relevant Paper: Autonomous UAVs as Means of Electronic Warfare

3.1 Bibliographic details

Title: “AUTONOMOUS UNMANNED AERIAL VEHICLES AS MEANS OF ELECTRONIC WARFARE TO PRESERVE THE LIVES OF PERSONNEL”

Authors: O. Kolomiitsev; O. Besspalko; O. Shmyhlenko.

Affiliation: Ivan Kozhedub Kharkiv National Air Force University.

Location in proceedings: printed page 547 (PDF pagination has an offset).

3.2 Core problem

The authors connect autonomous UAV-based EW with personnel protection. The proposed idea is to move part of the electronic-warfare sensing/action capability onto autonomous unmanned platforms so that the platform can observe hostile RF activity and operate with reduced continuous operator involvement.

3.3 RF information the proposed system would learn from

- Radar signals
- GPS signals
- Military communications
- Drone-control signals

3.4 Unknown-signal discovery: K-Means and DBSCAN

A key idea is self-learning identification/classification of previously unknown signals. The authors specifically name K-Means and DBSCAN as clustering approaches.

Technical interpretation: the receiver can represent observations using measurable signal features, then group observations according to similarity rather than requiring every emitter to be known beforehand. K-Means partitions observations into a chosen number of clusters; DBSCAN instead discovers dense groups and can mark isolated observations as noise/outliers. Neither algorithm by itself establishes the real-world identity of an emitter; clustering is a pattern-discovery step.

3.5 Detection/classification: SVM, CNN and LSTM

The paper names Support Vector Machines (SVM) and neural-network approaches including CNN and LSTM for hostile-frequency detection/classification.

Technique	Technical role	Relevance to SIH
SVM	Supervised classification of feature vectors	Possible baseline classifier
CNN	Pattern extraction/classification; can operate on signal representations such as spectrogram-like inputs	Useful for signal-pattern recognition
LSTM	Sequence/time-series modelling	Potentially useful for predicting temporal emitter activity

3.6 Learning from accumulated RF data

The authors propose that, as training information accumulates, neural networks can analyse changes in the RF environment and predict changes in adversary behaviour. The predicted changes are then intended to support selection of frequencies and jamming methods.

For your SIH problem, the important conceptual transition is: observation → learning → prediction → adaptive decision. However, the paper does not present a measured implementation of your exact receiver scan scheduler.

4. Second Key Paper: Construction of Electronic Interference Using Machine Learning

4.1 Bibliographic details

Title: “CONSTRUCTION OF ELECTRONIC INTERFERENCE USING MACHINE LEARNING”

Authors: O. Yula; V. Raksha; V. Lytovchenko; K. Kvitkin.

Affiliations include the State Scientific Research Institute of Armament and Military Equipment Testing and Certification and Ivan Kozhedub Kharkiv National Air Force University.

Location: immediately following the preceding EW-UAV paper in the proceedings.

4.2 ML + Ant Colony Optimization

This paper proposes combining machine learning with Ant Colony Optimization (ACO) for adaptive electronic interference. The ML component is described as analysing incoming information and predicting changes in enemy signal characteristics. ACO is used as the optimization/search component for adjusting parameters and allocating resources.

The conceptual separation is useful for your project: ML can estimate or predict the environment, while an optimization layer can choose an action under constraints. In your SIH context, the action is not necessarily interference; it can instead be the next receiver frequency/time allocation.

The paper discusses optimization of energy and spectral resources and prediction of tactical changes. It should be treated as a proposed adaptive methodology, not as evidence that a particular ML+ACO system was operationally deployed or benchmarked.

5. Adaptive Radar Target Selection Under Interference

Paper: “AIR TARGET SELECTION ADAPTIVE METHOD IN RADAR APPLICATION.” Authors: O. Karpenko, Candidate of Technical Sciences, Associate Professor; A. Kiiko, Candidate of Physical and Mathematical Sciences. Affiliation: Ivan Kozhedub Kharkiv National Air Force University.

The paper addresses hostile air-target selection and tracking under intensive interference. It proposes an adaptive radar target-selection method based on statistical characteristics and classification of passive interference.

Technical aspects reported

- Consideration of targets across a wide speed range, including highly manoeuvring aerial objects.
- Use of statistical characteristics and classification of passive interference in adaptive target selection.
- Consideration of transmitted-signal parameters affecting moving-target selection under interference.
- Research on electromagnetic-wave characteristics including wavelength and polarization.
- Consideration of pulse-burst repetition frequency.
- Resolution-volume considerations based on transmitted-pulse duration.
- Optimization of the antenna radiation pattern beam width at half-power points.
- Emphasis on defining physical characteristics and selective criteria for spectral analysis of received signals.

Why it matters to your SIH

This is not a smart spectrum scheduler, but it demonstrates the broader principle that sensing parameters and selection criteria can be adapted to the observed interference environment rather than kept completely fixed.

6. Adaptive Tracking / Motion-Model Research

The proceedings also contain research on models of modern air-object motion for synthesis of radio-engineering tracking algorithms. The relevant concept is adaptive/parallel filtering using multiple motion models so that the tracking algorithm can better accommodate changing target manoeuvres.

Technical relevance: the same general architecture appears in many adaptive sensing problems—maintain competing hypotheses, update them from measurements, and allocate computational/sensing effort according to the current evidence.

7. How These Conference Developments Map to Your SIH Problem

Conference development	What it solves	What it does NOT establish	SIH connection
K-Means / DBSCAN	Unknown-signal clustering	Does not identify an emitter automatically	Discover patterns without a complete prior library
SVM / CNN	Signal detection/classification	Does not decide scan order by itself	Signal understanding layer
LSTM / temporal modelling	Sequence/activity prediction	No demonstrated SIH scheduler	Predict future band activity
ML + ACO	Prediction + resource/parameter optimization	Paper is about electronic interference, not your receiver scheduler	Decision/optimization philosophy
Adaptive radar selection	Selection under interference	Not an RF sweep scheduler	Adaptive sensing principle
Adaptive tracking	Changing-target state estimation	Different sensing domain	Evidence for adaptive multi-model processing

8. Recommended Software Architecture for the SIH Prototype

A defensible software-only architecture inspired by the conference concepts—but specifically tailored to your SIH problem—would separate signal understanding from scheduling.

1. RF environment simulator: generate a time-frequency world with ground-truth emitter activity.
2. Receiver model: impose limited instantaneous bandwidth and a finite scan/retune cost.

3. Signal detector: decide whether useful activity is present in the currently observed band/time slot.
4. Feature extraction: convert observations into a compact feature vector.
5. Unknown-pattern module: use clustering to discover recurring signal behaviours.
6. Classifier/predictor: classify known patterns and estimate future activity probabilities.
7. Scheduler: choose the next band/time slot using predicted reward, uncertainty and scan cost.
8. Feedback: record hit/miss and update the model or scheduler.
9. Evaluation engine: compare the adaptive scheduler against random and round-robin baselines.

9. The Key Research Distinction

The conference material supports the feasibility and research relevance of adaptive RF analysis, unknown-signal clustering, ML-based detection/prediction and optimization. Your SIH problem adds a distinct constrained decision layer: the receiver cannot observe the whole spectrum at once and therefore has to decide where and when to spend its limited observation time.

A strong project thesis is therefore: “Use learned information about the RF environment to dynamically allocate limited receiver observation opportunities.” The novelty should be evaluated experimentally rather than asserted.

10. Metrics for Your Simulation

- Probability of Detection (Pd): fraction of relevant transmission opportunities successfully detected.
- Probability of False Alarm (Pfa): fraction of observations incorrectly treated as relevant activity.
- Average Intercept Time: average delay between an emitter becoming active and the receiver observing it.
- Average Intercept Rate: number/fraction of relevant opportunities successfully intercepted.
- Coverage: how much of the frequency-time search space is actually observed.
- Average reward/cost: a scalar objective combining detection benefit and scan/resource costs.
- Prediction accuracy: how accurately the model predicts future emitter activity.
- Intercept-time error: difference between predicted and actual useful interception time.
- Robustness: performance degradation when emitter behaviour, density, periodicity or frequency agility changes.

11. Baselines You Should Compare

Baseline	Decision rule	Purpose
Random scan	Choose next band randomly	Lower-bound/reference behaviour
Round-robin	Visit bands in fixed cyclic order	Classic open-loop baseline
Recency/probability heuristic	Prefer bands with recent/high activity	Simple adaptive benchmark
ML predictor + heuristic scheduler	Predict activity then rank bands	Shows value of prediction
Full adaptive scheduler	Prediction + uncertainty + reward/cost	Your proposed system

12. What the Conference Does NOT Prove

- It does not prove that Ukraine has fielded an operational K-Means/DBSCAN/CNN/LSTM receiver scheduler.
- It does not provide a validated benchmark showing that these methods outperform round-robin scanning.
- It does not provide a dataset for reproducing the proposed models.

- It does not provide Pd/Pfa/intercept-time results for the exact SIH problem.
- It does not establish that the described approaches are currently deployed in combat.
- The papers are conference research contributions/proposals; operational status must not be inferred from their presence in the proceedings.

13. Source and Citation

Primary source: Ministry of Defence of Ukraine / Ivan Kozhedub Kharkiv National Air Force University, XXI International Scientific Conference “New Technologies – for the Protection of Airspace,” 9–10 April 2025, 888 pages.

Official PDF: https://hups.mil.gov.ua/wp-content/uploads/admission-university/naukovij_central/XXI_naukova_konferencia_KhNUPS_2025.pdf

The conference front matter states that the proceedings contain abstracts of theoretical and practical research/development results and that responsibility for the factual content rests with the individual authors.

14. Bottom Line for the SIH Team

The strongest takeaway is not “Ukraine already solved this.” The defensible takeaway is that Ukrainian military research in 2025 publicly explored several building blocks that are directly relevant to your problem: self-learning RF analysis, unknown-signal clustering, ML-based detection/classification, temporal prediction, adaptive target selection, and ML + optimization for resource allocation. Your SIH problem can combine these ideas around the specific decision problem of allocating a bandwidth-limited receiver’s scan time across a changing frequency-time environment.